Chapter 12 & 13

PHARMACOEPIDEMIOLOGY IN DRUG SURVEILLANCE

DRUG SAFETY MONITORING

Reasons to perform PE studies

- Regulatory
  - To obtain approval for marketing
  - Response to question by regulatory agency
- Marketing
  - To assist market penetration by documenting the safety of the drug
  - To protect the drug from accusations about adverse effects
- Legal
  - anticipation of future product liability litigation
- Clinical
  - Hypothesis generating
  - Testing hypothesis

PE study contribution in drug surveillance

- General contributions of pharmacoepidemiology
  - Reassurances about drug safety
  - Fulfillment of ethical and legalobligations
- Information which supplements the information available from pre-marketing studies.
  - In patients not studied prior to marketing, e.g. the elderly, children, pregnant women
  - Effect modified by other drugs and other illnesses
- New types of information not available from premarketing studies.
  - Discovery of previously undetected adverse and beneficial effects.
  - Uncommon effects.
  - economic implications

Pharmacovigilance

- The word Pharmacovigilance derived from
  - Pharmacon : Drug
  - Vigilare : to be alert
- The science and activities relating to the detection, assessment, understanding, and prevention of adverse effects or any other drug-related problems.
- Adverse drug reaction: a response to a drug which is noxious and unintended which occur at dose used for prophylaxis, diagnostic or therapy of disease (causality directly associated with medicine).
- Adverse event: Causality not directly associated with medicine.
  - Anti-histaminics: Sedation causes accident and injury.
  - Alpha-blockers: Fall in BP- Fall and fracture

Pharmacovigilance: Aim & Objectives

- **Aim:**

To identifying new information about hazards as associated with medicines.

- **Objective:-**
- Improve patient care and safety
- Improve public health and safety
- Encourage safe, rational and appropriate use of drugs
- Promote understanding, education and clinical training in pharmacovigilance

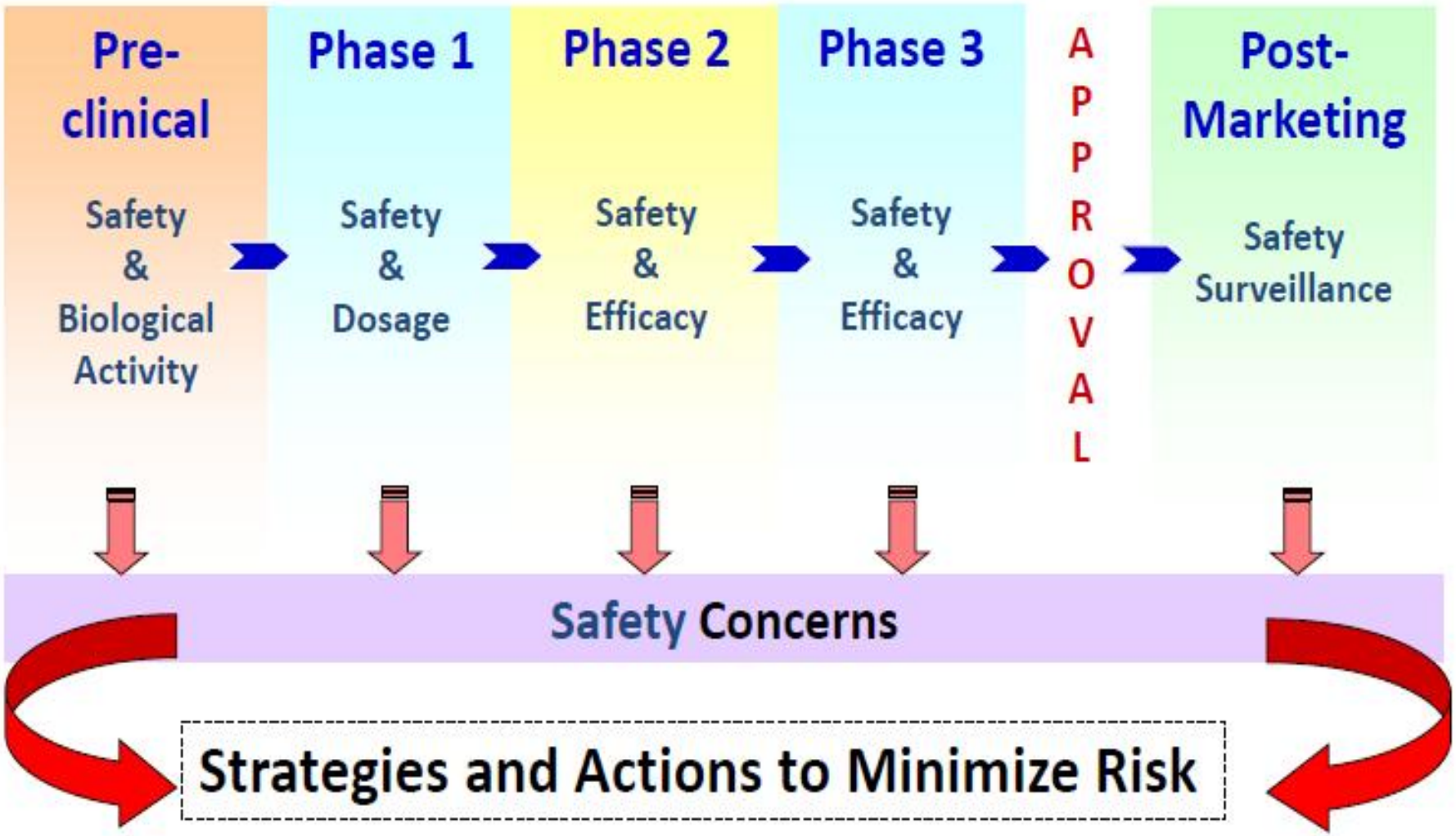
Key stakeholder in PV

- Patient
- Health professionals
- Marketing Authorization Holders
- Regulatory agencies

Pharmacovigilance

- Pre-authorization PV
  - Analysis of safety information (obtained from pre-clinical and Clinical studies) collected before the marketing authorization of the drug.
- Post-authorization PV
  - Analysis of safety information (obtained from spontaneous case reports, literature reports, safety studies) collected after the marketing authorization of the drug.

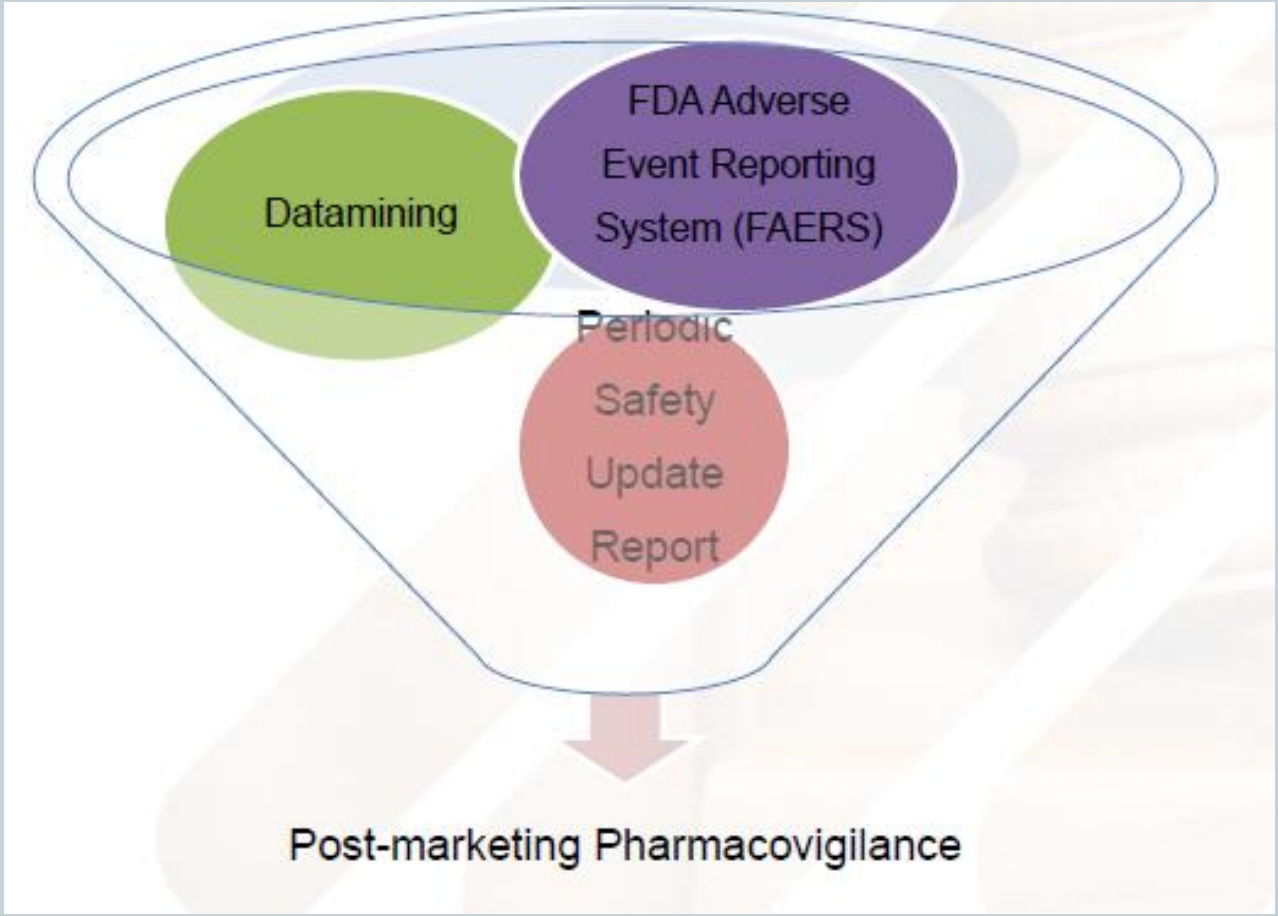
PHASES OF CLINICAL TRIALS



Limitation of Pre-approval clinical trial

- Trial population
  - Size
    - Trial population vs. treated population
  - Narrow
    - Very young or very old usually not enrolled
  - Co-morbidities
    - Hepatic or renal failure
    - Other serious medical conditions
    - Use of concomitant medications
- Indications for use
  - Proposed indication for use
    - Patients at complex disease stages often not enrolled
- Duration of trial
  - Typical chronic use (years) vs. trial (several weeks to months)

Routine Post-marketing Pharmacovigilance



POST-MARKETING SURVEILLANCE

- No fixed duration/Patient population.
- Starts immediately after marketing approval
- Report all ADRs
- Drug-drug interaction
- Drug-food interaction
- Misuse or abuse of drug
- Medication error (Product packaging, labeling etc)
- Help to detect
  - Rare ADRs
  - Drug interaction
  - Also new uses for drugs [sometimes called Phase V]

Sources of PE data

- Spontaneous AE reporting
- Global Drug surveillance: Uppsala Monitoring Centre, UMC
- Case- control surveillance
- Prescription event monitoring
- Automated databases
- Others

METHODS OF PMS

- Thus, four types of studies are generally used to identify drugs effects:
  - 1. Controlled clinical trials (RCT),
  - 2. Spontaneous or voluntary recording
  - 3. Cohort, studies and
  - 4. Case control studies.

Drug safety monitoring

- To market a drug, the manufacturer must provide evidence of its efficacy and safety to the U.S.FOOD and Drug Administration (FDA)
- In Pre-marketing study, the numbers and type of patient used to demonstrate a drug`s efficacy and safety are limited.
- Although pre-marketing surveillance cannot provide knowledge of the safety or efficacy of the drug at the time of introduction into the market.
- Post-marketing surveillance of drug therefore play an important role to discover an undesirable effect that might present at risk.
- It provide additional information on the benefit and risk of the drugs.

Adverse Drug Experience

- Any undesirable event that is associated with the use of a drug in humans, whether or not considered drug-related and occurs in the course of the use of a drug product in professional practice. This may include:
- Drug overdose
- Drug abuse
- Drug withdrawal
- Any failure of expected pharmacologic action

Serious Adverse Experience

- Death
- Life-threatening adverse experience
- Inpatient hospitalization
- Persistent/significant disability
- Congenital birth defect

Adverse drug reaction

	Type A (Augmented)	Type B (Bizarre)
Pharmacologically predictable	Yes	No
Dose dependent	Yes	No
Frequency	Common	Rare
Incidence	High	Low
Martality	Low	High
Treatment	Adjust dose	Stop drug

Spontaneous Reports and FAERS

FAERS

(FDA Adverse Event Reporting System)

In spite of its limitations (underreporting , incomplete reports, etc) spontaneous reports of AEs provide an important cornerstone for pharmacovigilance in the US.

Yellow Card Scheme:- The UK national spontaneous ADR reporting scheme.

FDA initiation of Labeling change such as a 'boxed warning'

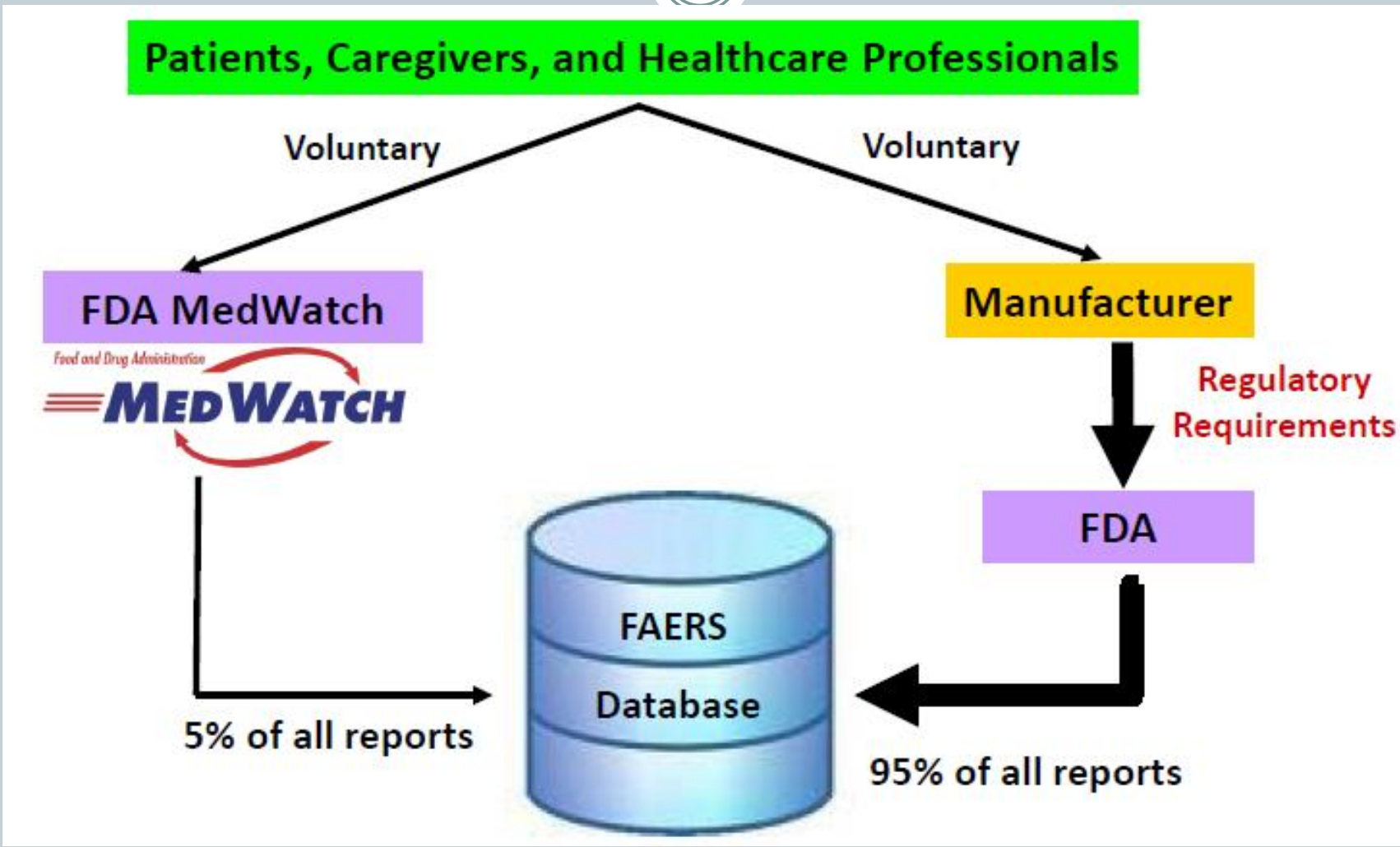
FDA Adverse Event Reporting System (FAERS)

- FAERS contains adverse drug reaction reports which FDA received from manufacturers as regulatory requirement.
- FAERS comprises of Single or multiple Individual Cases Safety Reports (ICSRs)
- The MedWatch also site provides information about voluntary and mandatory reporting of FAERS

FAERS

- Fully automated computerized database
- Spontaneous reports
- Contains human drug and therapeutic biologic reports
- ~14 million reports since 1969
- Over 1.81 million new reports in 2017

Post marketing Reports System to FDA



Signal v/s Spontaneous reporting v/s PE

- Signal identified via spontaneous reporting confirmed by a formal Pharmacoepidemiology study.
- Example
  - Phenylpropanolamine (PPA)- ingredient used in OTC and prescription cough and cold medications as a decongestant, and in OTC weight loss products.
  - In 1984, FDA received reports of Hemorrhagic stroke (bleeding into brain or into tissue surrounding brain) in association with PPA.
  - In addition, there were published reports in literature.
- Question
  - Is the use of PPA-containing products associated with hemorrhagic stroke?

PMS signal sources

- Passive surveillanceSpontaneous/voluntary reporting of cases
  - National (FDA MedWatch)
  - Local or Regional
  - Scientific literature publications
- Active surveillance
  - Drug-Induced Liver Injury Network
  - Sentinel initiative
- Post-marketing studies
  - Observational studies
  - Randomized clinical trials

Characterize the signal in a Drug Safety Review

Characterize the signal in a Safety Review	Standard scientific format to document
–Written communication –Summary and analysis of a specific safety issue – Often results in recommendations for regulatory action	–Abstract –Background –Methods & Materials –Results –Discussion –Conclusion –Recommendations

Regulatory Action



Note: REMS –Risk Evaluation & Mitigation Strategy, HCP –Health Care Provider, DSC –Drug Safety Communication

FDA Communications

- MedWatch Safety Alerts and Drug Safety Communications
- Potential Signals of Serious Risks/New Safety Information Identified from FAERS
- Published literature and scientific meetings
- Teleconferences with foreign regulatory agencies:
  - EMA: European Medicines Agency
  - International Post-Market Surveillance (IPMS): Canada, Australia, New Zealand, Switzerland, and Singapore (via written submissions)